آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre on HSSC-II Business Statistics Annual Examinations 2024

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC) Part II Business Statistics. It contains comments on candidates' responses to the 2024 HSSC-II Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses that support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfill the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

Most candidates achieved success in constructing good responses. Especially in the concepts of

- Construction of Frequency Distribution Table
- Drawing Histogram and Frequency Polygon
- Stem-Leaf Plotting
- Variance of the group data
- Coefficient of Correlation.

Nonetheless, it is essential for teachers to concentrate on the following content and provide candidates with more drills and practice to foster a solid understanding.

- Properties of Arithmetic Mean
- Price Index Numbers
- Probabilities

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS
Constructed Response Questions (CRQs)

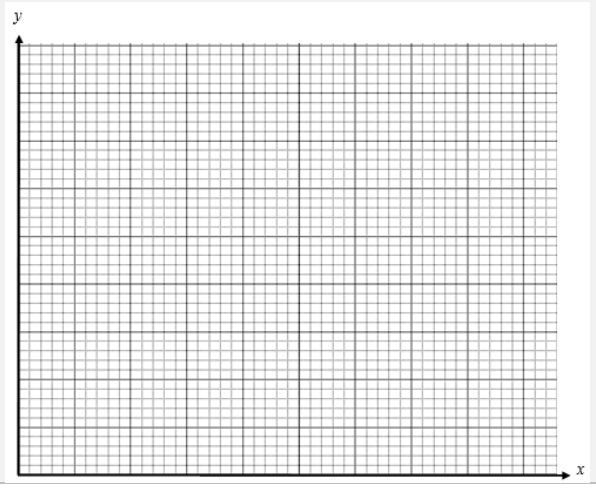
Question No. 1

Question Text

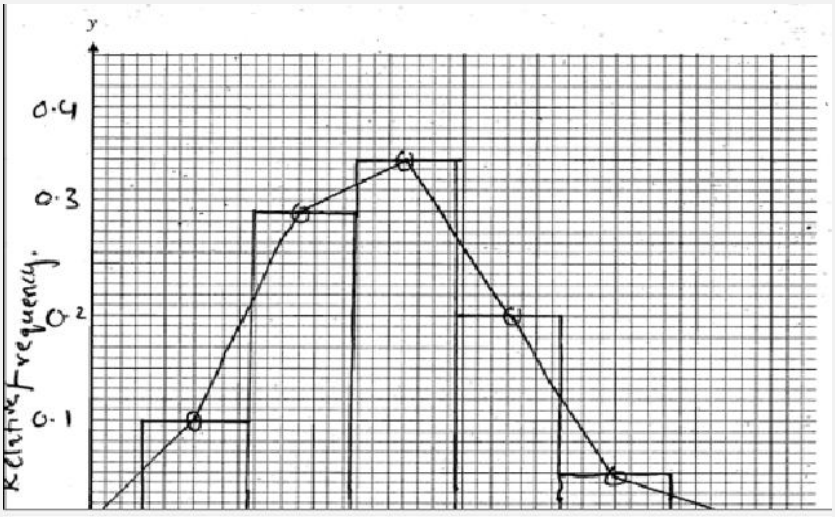
i. Complete the given frequency distribution table.

Marks	Relative Frequency	Class Mark
41.5 – 46.5	0.1	
46.5 – 51.5	0.3	
51.5 – 56.5	0.35	
56.5 – 61.5	0.2	
61.5 – 66.5	0.05	

ii. Construct the relative frequency histogram and relative frequency polygon for the data given in part (i).



SLO No.	10.1.1 and 10.1.5
SLO Text	Construct a frequency distribution table of discrete and continuous data including equal and unequal class intervals, tally marks/ frequency, upper class limit/ boundary, lower class limit/ boundary, class mark/ midpoint, class width and cumulative frequency. Draw a relative frequency histogram and polygon to extract information from graph.
Max Marks	3
Cognitive Level	A*
Checking Hints	1 mark for writing class marks 1 mark for making correct histogram 1 mark for making correct relative frequency polygon
Overall Performance	Overall, candidates performed well in this question. Candidates who performed well demonstrated a clear understanding of constructing and interpreting frequency distribution tables and graphs. They accurately represented the relative frequencies and plotted the data correctly. Weaker responses often involved errors in calculating and representing class marks and cumulative frequencies, leading to inaccuracies in the graphs.
Description of Better Responses	**Frequency Distribution Table:** Better responses correctly filled in the frequency distribution table, identifying and calculating class marks. They demonstrated an understanding of class boundaries and widths. **Relative Frequency Histogram and Polygon:** Better responses accurately constructed the histogram with relative frequencies on the y-axis and class boundaries on the *x*-axis. The relative frequency polygon was drawn by correctly plotting and connecting the midpoints of the histogram bars. Candidates used appropriate methods to ensure that the graphs were accurately represented.

Image of Better Response	**Part (i)** 	Marks	Relative Frequency	Class Mark
---	---	---
41.5 – 46.5	0.1	44
46.5 – 51.5	0.3	49
51.5 – 56.5	0.35	54
56.5 – 61.5	0.2	59
61.5 – 66.5	0.05	64	 **Part (ii)** 
Description of Weaker Responses	**Frequency Distribution Table:** Weaker responses struggled with identifying class marks and class boundaries, leading to errors in the table. Some of the candidates provided incorrect cumulative frequencies or misunderstood the concept of class width. **Relative Frequency Histogram and Polygon:** Many candidates either drew straight lines over the histogram bars or joined the corners of the histogram to form the polygon, which is incorrect. They failed to correctly plot the midpoints for the relative frequency polygon and had difficulties with accurate scaling and plotting on the graphs. Incorrect handling of class marks and frequencies led to a poorly constructed histogram and polygon.
Images of Weaker Responses	**Part (i)** 	Marks	Relative Frequency	Class Mark
---	---	---
41.5 – 46.5	0.1	41.5+46.5/2 = 44
46.5 – 51.5	0.3	46.5+51.5/2 = 49
51.5 – 56.5	0.35	51.5+56.5/2 = 54
56.5 – 61.5	0.2	56.5+61.5/2 = 59
61.5 – 66.5	0.05	61.5+66.5/2 = 64 | **Image (ii)** | Marks	Relative Frequency	Class Mark	c.f ≤	c.f >
---	---	---	---	---
41.5 – 46.5	0.1	10	10	100
46.5 – 51.5	0.3	30	40	90
51.5 – 56.5	0.35	35	75	60
56.5 – 61.5	0.2	20	95	25
61.5 – 66.5	0.05	5	100	5	 100

Part (ii) Image (i)

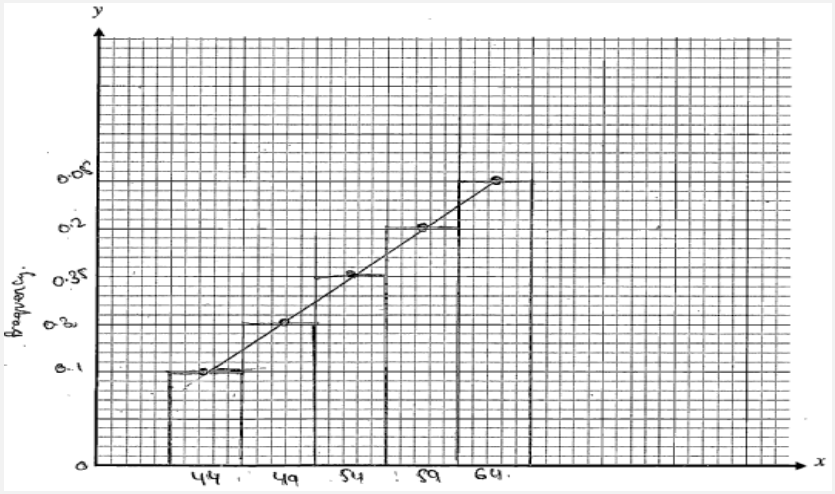


Image (ii)

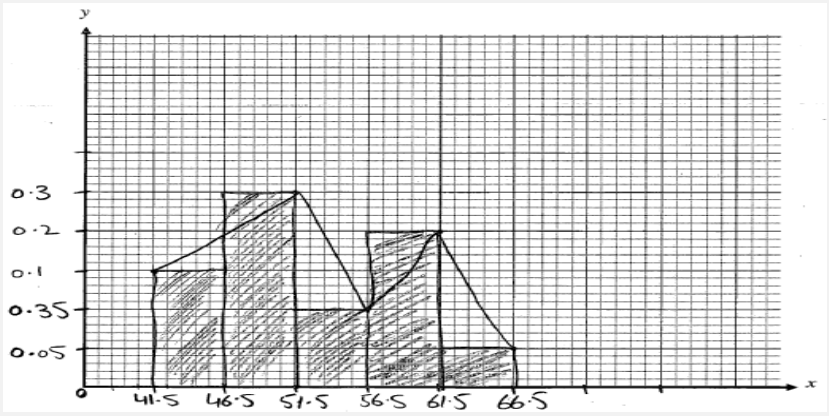
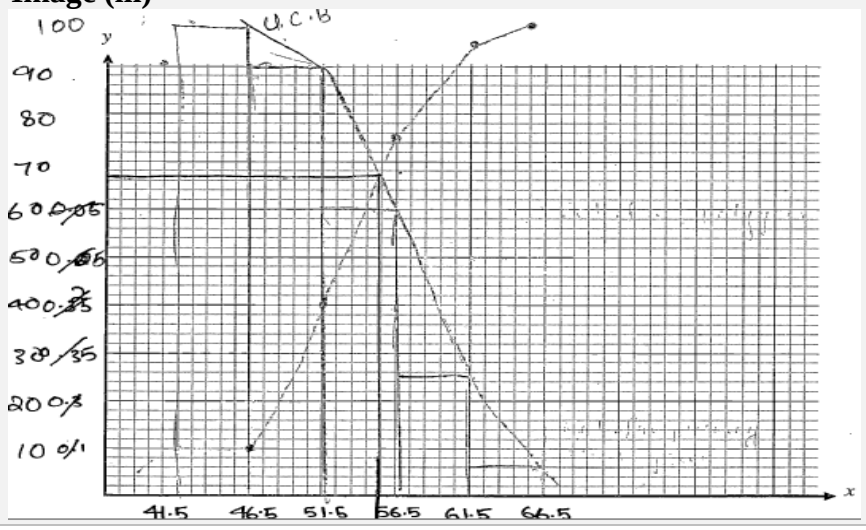


Image (iii)



Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy** Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide)	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

• Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each Pedagogy, refer to Annexure A	

Any Additional Suggestion:

To enhance the performance in this concept, teachers may consider the following strategies;

- **Digital Tools:** Utilise graphing software like GeoGebra or Excel to practice constructing and analysing frequency distributions and histograms. Interactive tools can help students visualise and understand the concepts more clearly.
- **Real-Life Examples:** Use real-life data sets, such as survey results or experimental data, to create frequency distribution tables and graphs. This approach can make the concepts more relatable and easier to understand.
- **Hands-On Activities:** Conduct in-class activities where students manually create frequency tables and draw histograms and polygons using provided data. This hands-on practice reinforces learning and builds confidence.
- **Practice Problems:** Assign diverse practice problems involving different types of data sets and class intervals to build proficiency in constructing and interpreting frequency distributions and graphs.

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

Question No. 2a **Candidates were given the choice to attempt any ONE out of the two questions: 2a and 2b.**	
Question Text	The average number of ATM transactions per day of a particular bank were recorded at 10 different locations in a city. The data were: 35, 50, 225, 50, 30, 65, 40, 55, 52 and 76. i. Find the mean of average number of ATM transactions per day. ii. On a certain day, due to a technical fault the number of ATM transactions were decreased by half from all of the locations. After technical fault, find the new mean of average number of ATM transactions. iii. Prove the following property: Half of the average number of ATM transactions before technical fault = Average number of ATM transactions after technical fault.
SLO No.	11.1.3(d)
SLO Text	Apply the following properties of arithmetic mean to solve related problems if $Y = \frac{X}{a}$, then $\bar{Y} = \frac{\bar{X}}{a}$ *or* $\bar{Y} = \frac{1}{a}(\bar{X})$
Max Marks	3
Cognitive Level	A

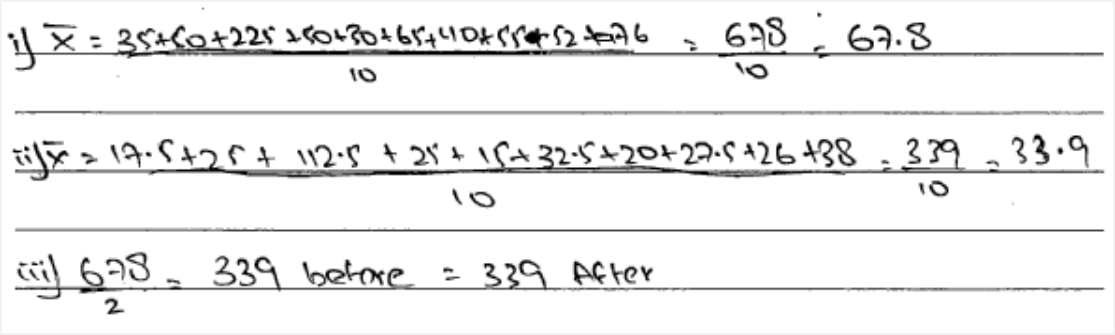
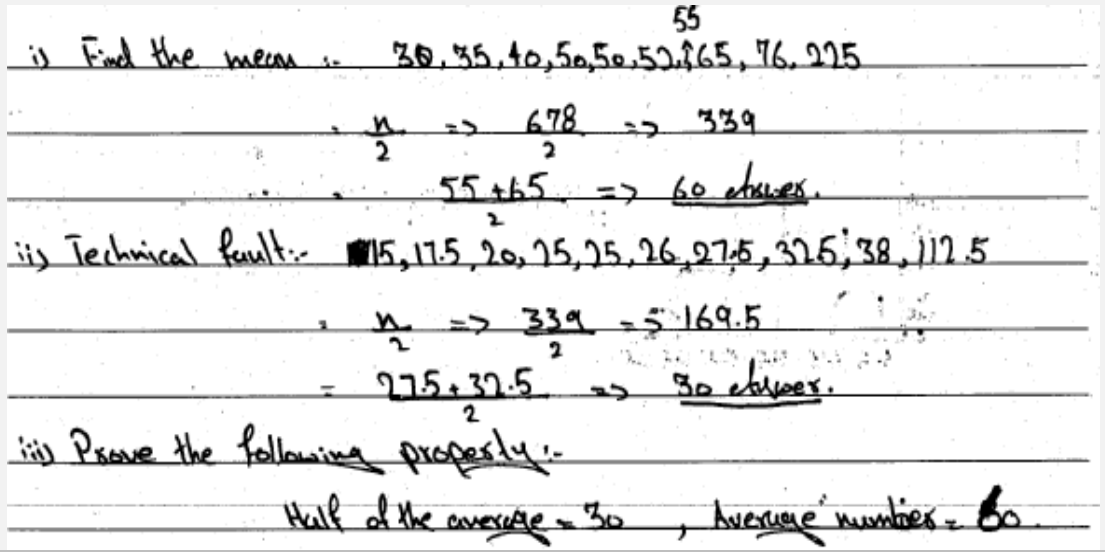
Checking Hints	1 mark for $\overline{X}$ 1 mark for $\overline{Y}$ 1 mark for $\overline{Y} = \frac{\overline{X}}{2}$
Overall Performance	Majority of candidates attempted this question. Candidates generally showed a good understanding of calculating the arithmetic mean and applying it to different situations. The most common issue was in handling the technical fault scenario, where candidates struggled with the concept of halving each data point rather than just adjusting the mean.
Description of Better Responses	**Part (i):** Better responses correctly calculated the mean by summing all transaction values and dividing by the total number of locations. **Part (ii):** Better responses accurately found the new mean by first halving each number of transaction values and then calculating the mean of these new values. They understood that the mean value needs to reflect changes applied to all data points, not just the previously computed mean. **Part (iii):** Better responses correctly demonstrated the property by showing that half of the initial mean is equal to the new mean, aligning with the expected result.
Image of Better Response	
Description of Weaker Responses	**Part (i):** Weaker responses generally performed well in finding the mean, with minimal issues. **Part (ii):** Weaker responses mistakenly divided the previously obtained mean by 2 instead of halving each individual value of number of transactions first and then recalculating the mean. This error indicated a misunderstanding related to the impact of change in values on the mean. **Part (iii):** Weaker responses often failed to prove the property correctly due to errors in part (ii) or misunderstanding of the relationship between the means before and after the technical fault.
Images of Weaker Responses	**Image (i)** 

Image (ii)

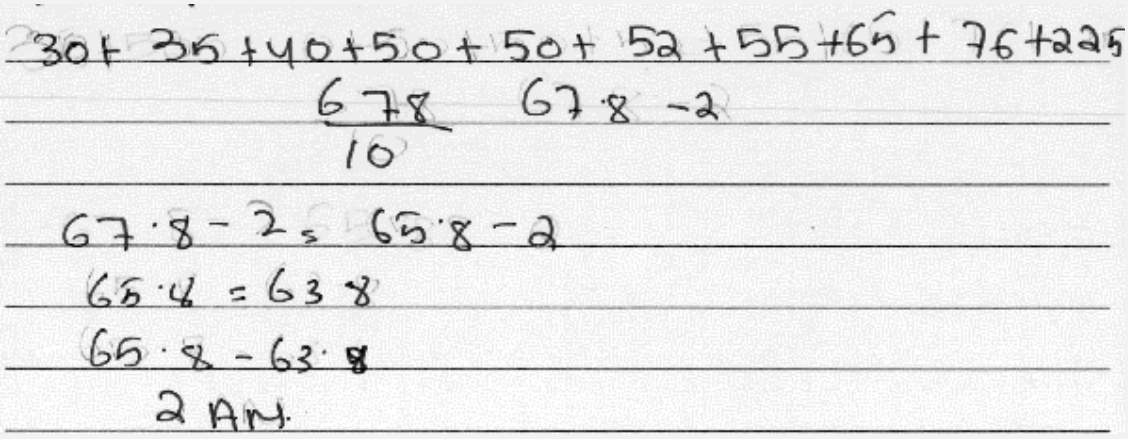
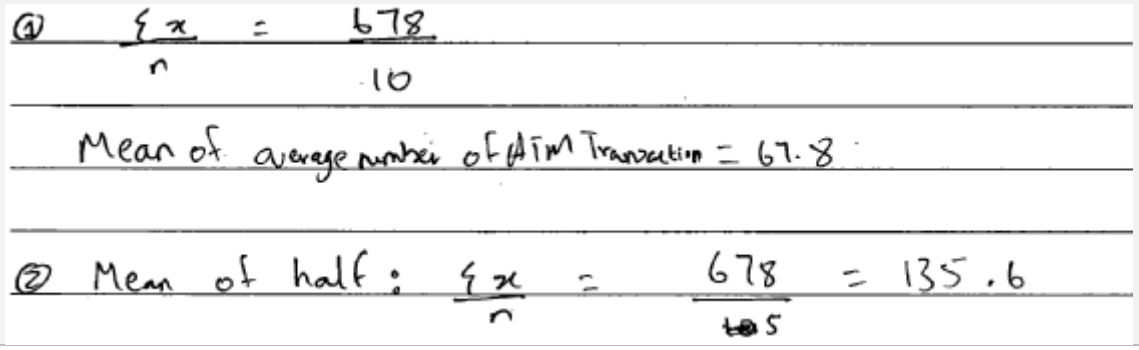


Image (iii)



Suggestions for improvement (Highlight all that apply)

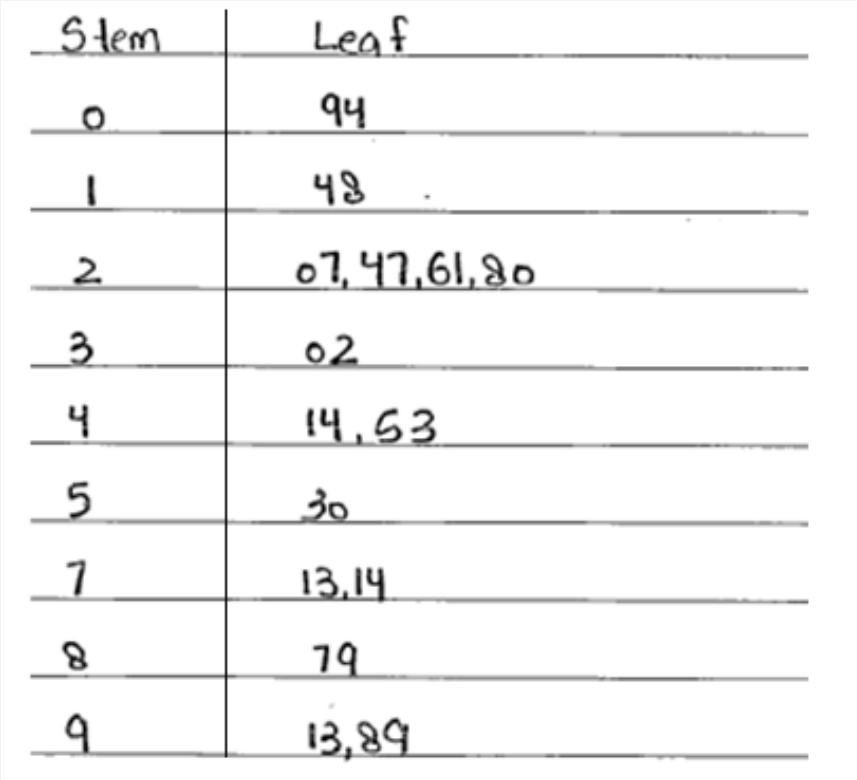
Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

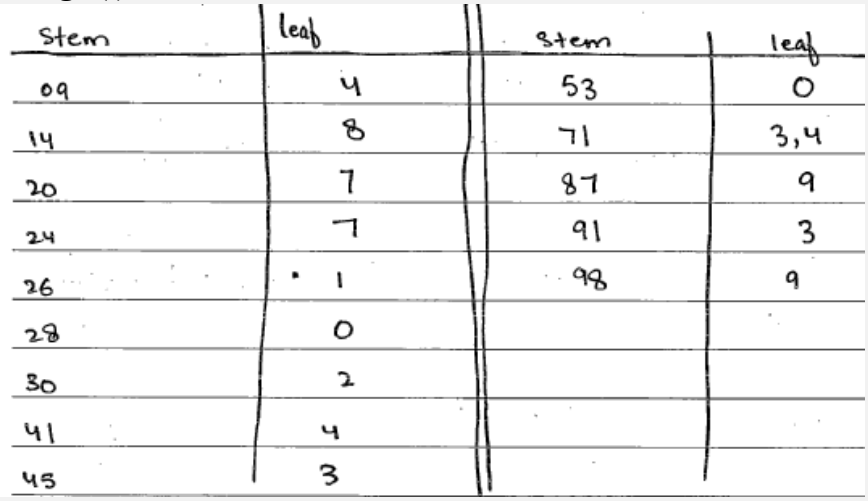
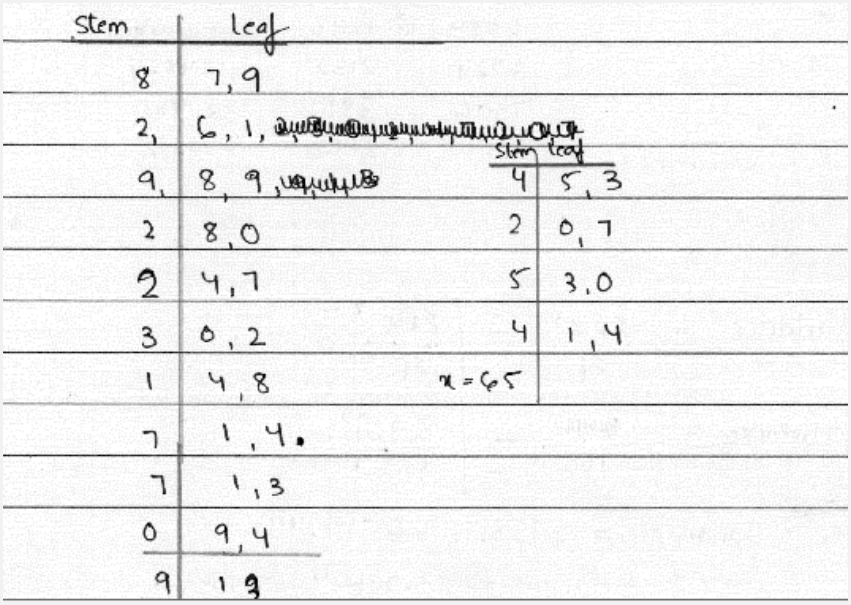
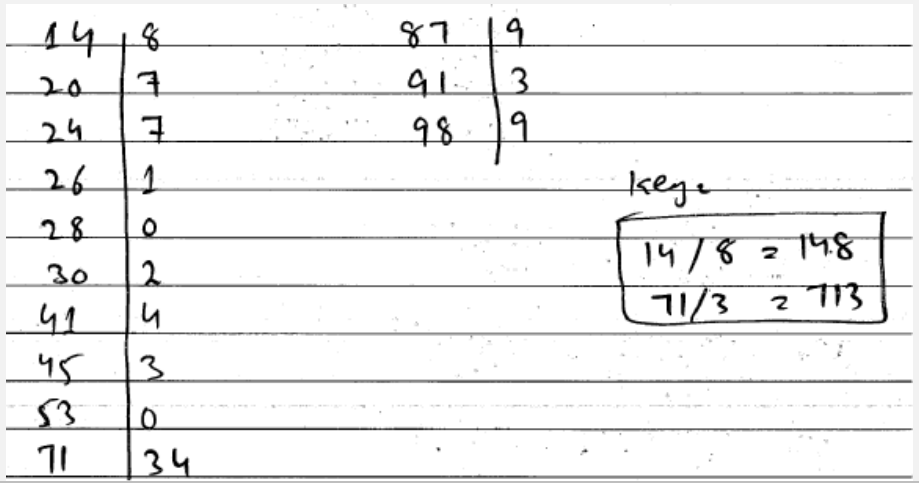
Any Additional Suggestion:

To enhance the performance of the candidates, teachers may consider the following strategies;

- **Digital Tools:** Use software like Excel or Google Sheets to practice calculations of mean before and after data adjustments. Visual tools can help students understand the impact of changes in data points on the mean.
- **Real-Life Examples:** Provide real-world scenarios involving average calculations and adjustments, such as changes in test scores or temperature readings, to make the concept more relatable.
- **Conceptual Understanding:** Emphasise the difference between adjusting individual data points and adjusting the mean value itself. Use diagrams and step-by-step explanations to clarify this concept.

- **Interactive Exercises:** Create exercises where students can manually adjust data points and recalculate the mean, reinforcing the impact of changes on the average value.
- **Practice Problems:** Offer a range of problems involving changes in data and mean calculations to build proficiency and confidence in handling various situations.

Question No. 2b **Candidates were given the choice to attempt any ONE out of the two questions: 2a and 2b.**	
Question Text	Usually the flow of river is measured in cubic metre per second (m^3/s). The following are measurements of a river's annual maximum flow: 879, 261, 280, 989, 247, 302, 148, 714, 713, 094, 913, 453, 207, 530, 414 Construct a stem-leaf plot with two-digit leaves.
SLO No.	11.3.5(a)
SLO Text	Construct stem and leaf display/ plot to summarise data.
Max Marks	3
Cognitive Level	U
Checking Hints	1 mark for identifying the correct stem from the question. 1 mark for putting the correct value in the leaf column. 1 mark for plotting the summarised data in table in correct order.
Overall Performance	Lesser number of candidates attempted this part. Candidates who performed well showed a good understanding of constructing stem-and-leaf plots, including proper data arrangement and accurate plotting. Weaker responses often involved a misunderstanding of the plotting process and incorrect placement of data.
Description of Better Responses	Better responses accurately created a stem-and-leaf plot by first arranging the data in ascending order. They used the tens digits as the stems and the unit's digits as the leaves. Such candidates organised their plot effectively, ensuring that each stem was correctly listed with corresponding leaves. They demonstrated a clear understanding of presenting data in a stem-and-leaf plot, making it easy to interpret.
Image of Better Response	

Description of Weaker Responses	**Incorrect Plot Construction:** Weaker responses misunderstood the requirement and did not arrange the data in ascending order before constructing the plot. Some of them placed the entire value in the stem column or confused the stem and leaf positions. **Data Presentation Errors:** Weaker responses failed to separate the data correctly, placing entire values in the stem or not properly dividing tens and units. This led to errors in their stem-and-leaf plots, making them difficult to interpret the data in stem-leaf plot.
Images of Weaker Responses	**Image (i)**  **Image (ii)**  **Image (iii)** 

Stem	leaf		Stem	leaf
09	4		53	0
14	8		71	3,4
20	7		87	9
24	7		91	3
26	1		98	9
28	0			
30	2			
41	4			
45	3			

Stem	leaf
8	7, 9
2,	6, 1,
9,	8, 9,
2	8, 0
2	4, 7
3	0, 2
1	4, 8
7	1, 4.
7	1, 3
0	9, 4
9	1 9

stem	leaf
4	5, 3
2	0, 7
5	3, 0
4	1, 4

$x = 65$

14	8	87	9
20	7	91	3
24	7	98	9
26	1		
28	0		
30	2		
41	4		
45	3		
53	0		
71	34		

key:

14 / 8 = 148

71/3 = 713

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

To enhance the performance of the candidates, teachers may consider the following strategies;

Digital Tools:

- **Online Plotting Tools:** Use interactive tools like Google Sheets or online stem-and-leaf plot generators to practice creating and interpreting stem-and-leaf plots. These tools can help visualise the data and reinforce correct plotting techniques.
- **Simulation Software:** Utilise software that simulates data sorting and plotting to demonstrate the correct construction of stem-and-leaf plots.

Real-Life Examples:

- **Practical Applications:** Use real-world examples, such as survey results or temperature readings, to construct stem-and-leaf plots. This makes the concept more relatable and helps students see its practical uses.
- **Classroom Activities:** Conduct activities where students collect and plot data from everyday situations, such as test scores or heights of students, to practice creating stem-and-leaf plots.
- **Visual Aids:** Provide diagrams and step-by-step instructions on creating a stem-and-leaf plot. Emphasise the importance of sorting data and correctly placing stems and leaves.
- **Group Discussions:** Facilitate group discussions on common mistakes and different ways of avoiding them. Encourage students to share their plots with peers/ groups and discuss discrepancies to improve understanding.

Question No. 3

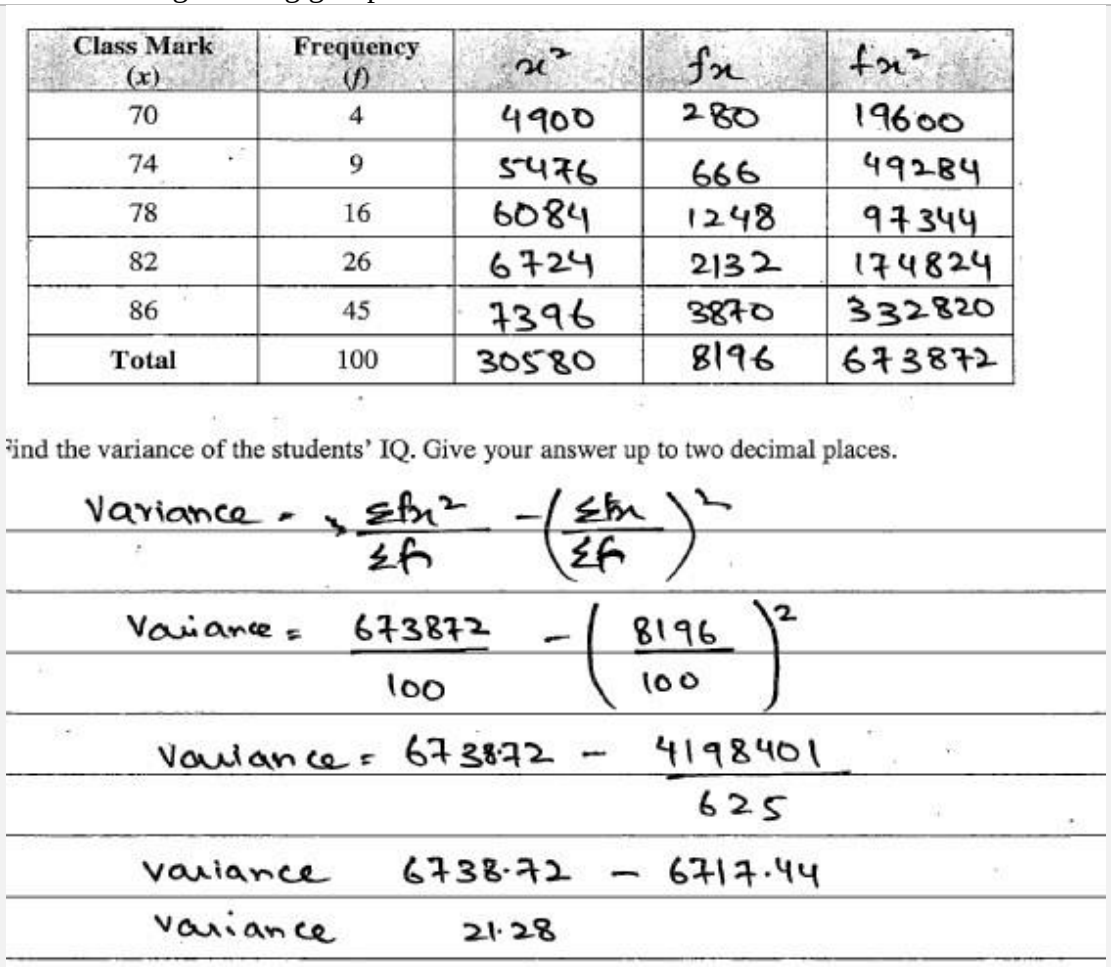
Question Text

The IQs of 100 students at a certain elementary level school are given in the following table.

Class Mark (x)	Frequency (f)			
70	4			
74	9			
78	16			
82	26			
86	45			
Total	100			

Find the variance of the students' IQ. Give your answer up to two decimal places.

SLO No. 12.1.4(a)

SLO Text	Calculate variance and standard deviation for grouped data by using the formulae. $\text{variance}: \sigma^2 = \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2 \text{ or } \sigma^2 = \frac{\sum f(x-\bar{x})^2}{\sum f}$
Max Marks	3
Cognitive Level	A
Checking Hints	1 mark for finding the values of $\sum fx$ or $\overline{X}$ 1 mark for finding the values of $\sum fx^2$ or $\sum(x-\overline{X})$ 1 mark for correct calculation of the variance
Overall Performance	Majority of the candidates performed well and demonstrated a clear understanding of calculating variance for grouped data, accurately applying the correct formula and values. Weaker responses revealed a misunderstanding of the variance formula specific to grouped data and incorrect application of variables.
Description of Better Responses	**Correct Calculation:** Better responses accurately computed x^2, $f(x)$, and $f(x^2)$ values. **Accurate Application:** Candidates showed a clear understanding of the formula for variance, correctly applying it to the grouped data. They applied the formula correctly and substituted all values appropriately, leading to finding the correct variance. Their understanding of using grouped data in variance calculations was evident.
Image of Better Response	
Description of Weaker Responses	**Incorrect Formula Application:** Many candidates applied the formula for ungrouped data rather than the formula for grouped data. This error indicated a lack of understanding of the specific requirements for calculating variance with grouped data. **Misunderstanding of Variables:** Candidates often confused the variables $\bar{x}$, x, $f(x)$, and $f(x^2)$. They used incorrect values in the formula, leading to inaccurate variance calculations.

Class Mark (x)	Frequency (f)	x^2	fx	fx^2
70	4	4900	280	19600
74	9	5476	666	49284
78	16	6084	1248	97344
82	26	6724	2132	174824
86	45	7396	3870	332820
Total	100	30580	8196	673872

Find the variance of the students' IQ. Give your answer up to two decimal places.

Variance = $\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2$

Variance = $\frac{673872}{100} - \left(\frac{8196}{100}\right)^2$

Variance = $673872 - \frac{4198401}{625}$

Variance $6738.72 - 6717.44$

Variance 21.28

Images of Weaker Responses

Image (i)

Class Mark (x)	Frequency (f)	fx	fx^2	R.f
70	4	280	78,400	4
74	9	666	443,556	9
78	16	1248	1,557,504	16
82	26	2132	4,545,424	26
86	45	3870	14,976,900	45
Total	100	8196	21,601,784	-

Find the variance of the students' IQ. Give your answer up to two decimal places.

$$= \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2$$

$$= \frac{21,601,784}{100} - \left(\frac{8196}{100}\right)^2$$

$$= 216017.84 - 6717.4416$$

$$= 209300.39$$

Image (ii)

Class Mark (x)	Frequency (f)	x^2		
70	4	4900		
74	9	5476		
78	16	6084		
82	26	6724		
86	45	7396		
Total 390	100	$\Sigma x^2 = 30,580$		

Find the variance of the students' IQ. Give your answer up to two decimal places.

$$\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2$$

$$= \frac{30,580}{100} - \left(\frac{390}{100}\right)^2$$

$$= \frac{30,580}{100} - 15.21$$

$$= 290.59 \rightarrow \text{Variance}$$

Image (iii)

Class Mark (x)	Frequency (f)	C. frequency		
70	4	4		
74	9	13		
78	16	29		
82	26	55		
86	45	100		
Total	100	100		

Find the variance of the students' IQ. Give your answer up to two decimal places.

$\frac{\Sigma(x-\bar{x})^2}{n}$

$\frac{\Sigma(86-26)^2}{100}$

$\frac{\Sigma(60)^2}{100}$

$\frac{\Sigma 3600}{100}$

$\frac{\Sigma 3600}{100}$

$\Sigma 36$ Ans.

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Pedagogy Used for that SLO**	**Assessment Strategies**
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Any Additional Suggestion:

To enhance the performance of the candidates, teachers may consider the following strategies;

Digital Tools:

- **Statistical Software:** Utilise tools like Excel or statistical software to practice variance calculations. These tools can automate some steps and provide immediate feedback, helping students understand the process.
- **Interactive Simulations:** Use online calculators or interactive platforms to illustrate the steps in calculating variance for grouped data.

Real-Life Examples:

- **Practical Data Sets:** Provide examples from real-life data, such as test scores or survey results, to illustrate the calculation of variance. This can help students see the relevance and application of the formula.
- **Classroom Activities:** Conduct activities where students calculate variance for different data sets, including both grouped and ungrouped data, to reinforce the correct methods.

Question No. 4	
Question Text	For the given data in the table, calculate Paasche's index number using base year 2010. Also interpret your result.

	2010		2015			
Commodity	**Quantity (kg)**	**Price/ kg (Rs)**	**Quantity (kg)**	**Price/ kg (Rs)**		
Wheat	100	10	120	13		
Rice	50	100	40	110		
Cotton	35	148	40	160		
Butter	12	600	10	750		
Mutton	48	500	35	600		
Total	-	-	-	-		

SLO No.	13.3.1
SLO Text	Calculate weighted price index by Laspeyeres, Paasche's and Fisher's indices.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for finding $p_o \times q_n$ 1 mark for finding $p_n \times q_n$ 1 mark for finding $\sum p_n \times q_n$ and $\sum p_o \times q_n$ 1 mark for finding correct value of index number and interpretation
Overall Performance	Overall candidates struggled in this question. Candidates who performed well showed a strong understanding of calculating and differentiating between Laspeyres, Paasche and Fisher's price indices. They correctly applied the formulas and handled the data accurately. Weaker responses indicated a misunderstanding of the indices and errors in data handling, leading to incorrect results.
Description of Better Responses	**Correct Calculation:** Better responses accurately applied the formulas for Paasche indices. They demonstrated a clear understanding of using base year and current year prices and quantities. **Proper Formula Application:** Most candidates correctly substituted the appropriate values into the formulas and performed calculations without errors. They displayed a strong grasp of the differences between the indices.

Image of Better Response

Commodity	2010 Quantity (kg) Q_0	2010 Price/ kg (Rs) P_0	2015 Quantity (kg) Q_n	2015 Price/ kg (Rs) P_n	$P_n Q_n$	$P_0 Q_n$
Wheat	100	10	120	13	1560	1200
Rice	50	100	40	110	4400	4000
Cotton	35	148	40	160	6400	5920
Butter	12	600	10	750	7500	6000
Mutton	48	500	35	600	21000	17500
Total	–	–	–	–	40860	34620

Paasche = $\frac{P_n \times Q_n}{P_0 \times Q_n} \times 100 = \frac{40860}{34620} \times 100$

$1.18024 \times 100 = 118.024\%$

The base year 2010 would have an index number of 100%. The Current year that is 2015 has an index number 118.0%. It shows that the index number of the current number has been increased by 18%. It means that the prices and Quantity in 2015 has increased by 18%.

Description of Weaker Responses

Misunderstanding of Indices: Many candidates confused the formulas for Laspeyres and Paasche indices.

Errors in Data Handling: Some of the candidates made mistakes in using current year prices, base year prices, or quantities. This confusion led to incorrect application of formulas and ultimately incorrect results.

Images of Weaker Responses

Image (i)

Commodity	2010 Quantity (kg)	2010 Price/ kg (Rs)	2015 Quantity (kg)	2015 Price/ kg (Rs)	$P_{n1} P_{n0}$	$P_{n0} P_{01}$
Wheat	100	10	120	13	12000	130
Rice	50	100	40	110	2000	11000
Cotton	35	148	40	160	1400	23680
Butter	12	600	10	750	120.	1350
Mutton	48	500	35	600	1680	30000
Total	–	–	–	–	17200	66169

$P_{on} = \frac{\Sigma P_n q_0}{\Sigma P_0 q_0} \times 100$

$\therefore P_{on} = \frac{\Sigma 17200}{66169} \times 100$

$\therefore P_{on} = 0.2599 \times 100$

$\therefore P_{on} = 25.997$ Answer

Image (ii)

Commodity	2010 Quantity (kg)	2010 Price/ kg (Rs)	2015 Quantity (kg)	2015 Price/ kg (Rs)	$P_n Q_0$	$P_0 Q_0$
Wheat	100	10	120	13	1300	1000
Rice	50	100	40	110	5500	5000
Cotton	35	148	40	160	5600	5180
Butter	12	600	10	750	9000	7200
Mutton	48	500	35	600	28800	24000
Total	–	–	–	–	50200	42380

$$\text{Paasche's} = \frac{P_n Q_0}{P_0 Q_0} \times 100$$

$$= \frac{50200}{42380} \times 100 = 118.45$$

Ans

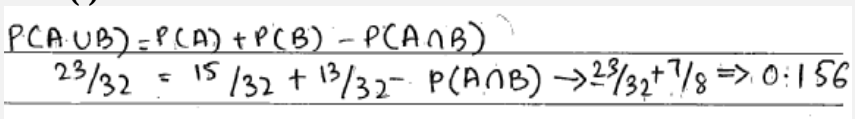
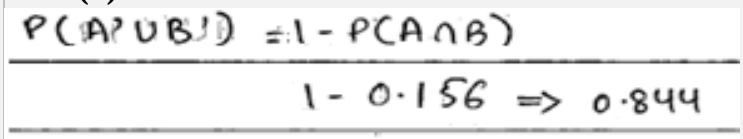
Suggestions for improvement (Highlight all that apply)

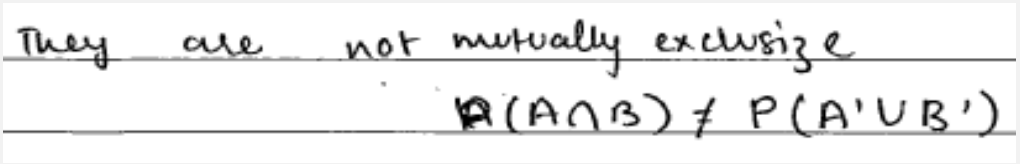
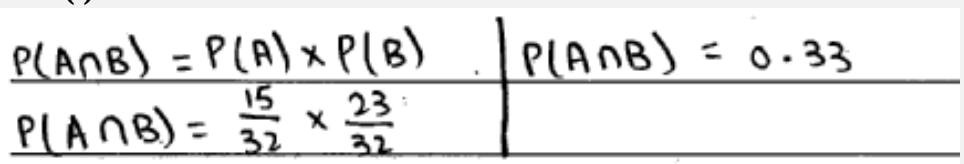
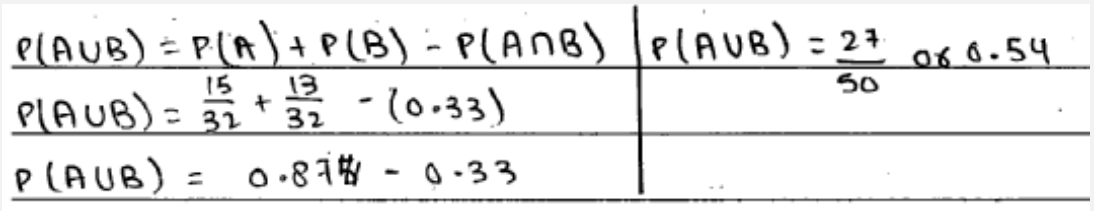
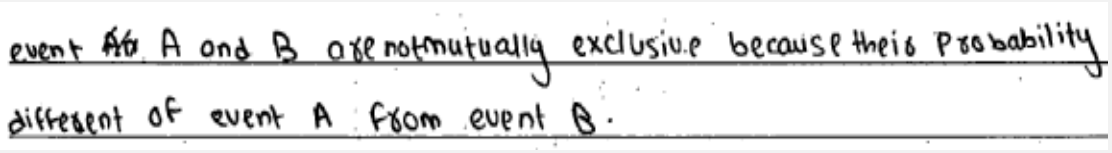
Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

To enhance the performance of this concept teachers may consider the following strategies;

- **Online Calculators:** Use online calculators or software for computing price indices to provide students with a clear example of applying the formulas. This can help them check their work and understand the correct process.
- **Interactive Simulations:** Implement simulations or interactive tools that allow students to input various values and see the change in indices, reinforcing their understanding of each formula.
- **Case Studies:** Provide real-world scenarios, such as changes in consumer prices over time, to help students understand the application of Laspeyres, Paasche and Fisher's indices in practical contexts.
- **Data Analysis Activities:** Use actual market data to create practice problems. Have students calculate indices based on this data to see the relevance and application of the formulas.

Question No. 5a Candidates were given the choice to attempt any ONE out of the two questions: 5a and 5b.	
Question Text	A and B are two events such that $P(A) = \frac{15}{32}$, $P(B) = \frac{13}{32}$ and $P(A \text{ or } B) = \frac{23}{32}$. i. Find $P(A \cap B)$ ii. Find $P(A' \cup B')$ iii. State whether A and B are mutually exclusive or not. Also justify your answer.
SLO No.	14.4.12
SLO Text	Apply the laws mentioned in SLOs 14.4.5, 14.4.7 and 14.4.10 to solve related problems. (14.4.5.b mutually non-exclusive i.e. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$).
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for finding $P(A \cap B)$ 1 mark for identity $P(A' \cup B') = 1 - P(A \cap B)$ 1 mark for finding $P(A' \cup B')$ 1 mark for identifying $P(A \cup B) \neq P(A) + P(B)$
Overall Performance	Majority of the candidates attempted this part as compared to part b. Candidates who performed well demonstrated a strong understanding of probability concepts and correctly applied the relevant formulas. They accurately calculated intersections, unions of complements and correctly identified the mutual exclusivity of events. Weaker responses showed a misunderstanding of probability laws, particularly the concept of mutual exclusivity and the correct application of formulae.
Description of Better Responses	**Intersection:** Better responses correctly used the formula $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ to find the intersection. They substituted the given values accurately. $P(A \cap B) = 15/32 + 13/32 - 23/32 = 5/32$. This demonstrates a strong grasp of applying the principle of inclusion and exclusion to find the probability of both events occurring together. **Union of Complements $P(A' \cup B')$:** Better responses correctly calculated $P(A' \cup B')$ using the formula $P(A' \cup B') = 1 - P(A \cap B)$. They understood that finding the probability of the union of complements involves first determining the intersection and then subtracting from 1. Their results reflected accurate calculations and understanding of handling complements and their unions. **Mutual Exclusivity:** Better responses accurately concluded that A and B are not mutually exclusive. They correctly noted that mutual exclusivity would imply $P(A \cap B) = 0$, however, here $P(A \cap B) = 532 \neq 0$, so, A and B cannot be mutually exclusive. This shows a clear understanding of the concept, justifying their conclusions based on the given probabilities.
Image of Better Response	**Part(i)**  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $23/32 = 15/32 + 13/32 - P(A \cap B) \rightarrow 28/32 + 7/8 \Rightarrow 0.156$ **Part(ii)**  $P(A' \cup B') = 1 - P(A \cap B)$ $1 - 0.156 \Rightarrow 0.844$

	Part(iii) 
Description of Weaker Responses	**Intersection $P(A \cap B)$:** Weaker responses struggled with the formula for intersection. Some used incorrect methods or confused the principles behind calculating $P(A \cap B)$. Common errors included applying the formula for independent events or mistakenly using the values in incorrect combinations. These errors led to incorrect answers and indicated a lack of understanding of correctly applying the formula for intersections. **Union of Complements $P(A' \cup B')$:** Weaker responses who made errors in calculating $P(A' \cup B')$ often also made mistakes in finding $P(A' \cup B')$. Some candidates incorrectly divided the mean or used irrelevant calculations, showing a misunderstanding about calculating the union of complements. They failed to connect the correct computation of intersections with the proper formula for the union of complements. **Mutual Exclusivity:** Weaker responses often included incorrect assumptions about mutual exclusivity. Some candidates mistakenly applied the concept of independent events or incorrectly deduced that events must be mutually exclusive because they did not understand the correct relationship between the probabilities. These misconceptions led to incorrect conclusions about the nature of *A* and *B*.
Image of Weaker Response	**Part(i)**  **Part(ii)**  **Part(iii)** 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide)	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform

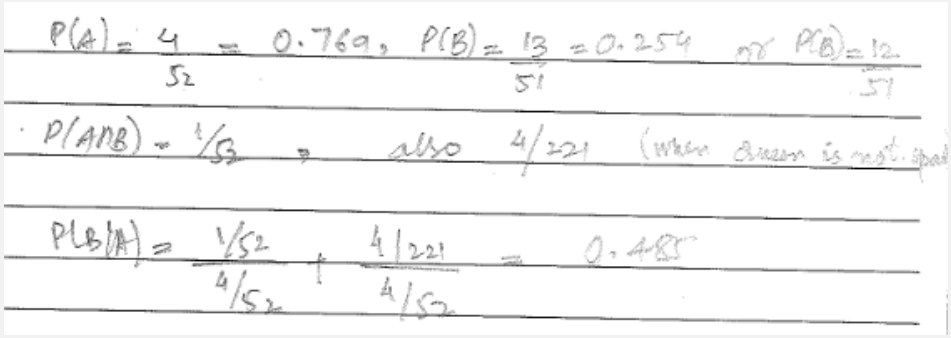
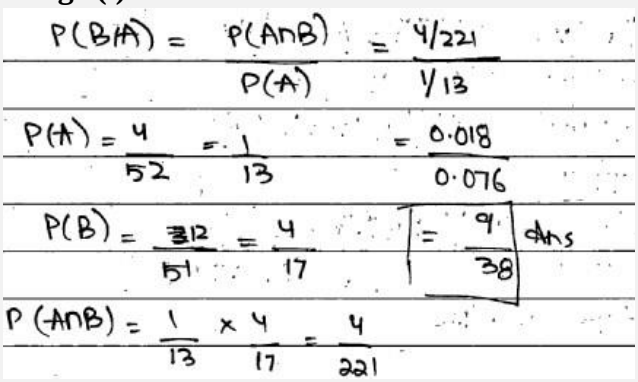
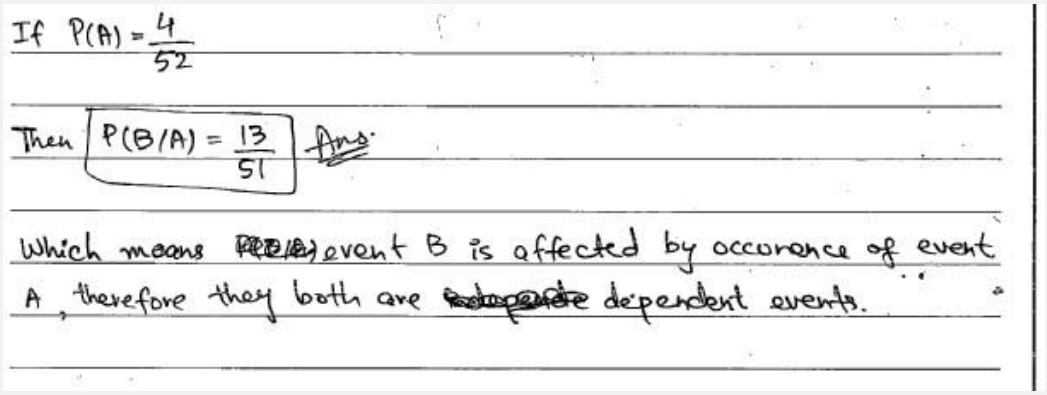
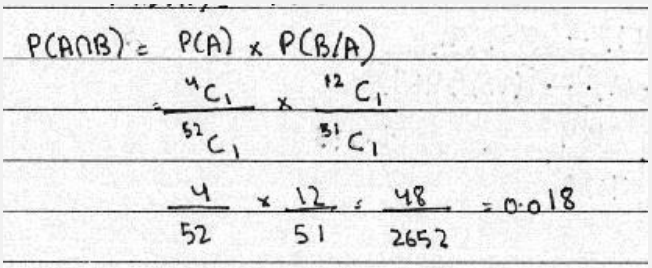
• Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

To enhance the performance of this concept, teachers may consider these strategies;

- **Interactive Probability Simulations:** Utilise online tools to visually demonstrate different ways in which probabilities interact, helping students grasp the concepts of intersections, unions and complements.
- **Probability Calculators:** Incorporate probability calculators into lessons to enable students to verify their answers and understand the step-by-step process of applying probability formulas.
- **Practical Scenarios:** Use real-world examples, such as probability problems involving card games or survey data, to make abstract concepts more tangible.
- **Classroom Activities:** Engage students with activities that involve calculating probabilities based on actual data, reinforcing the relevance of these concepts.

Question No. 5b **Candidates were given the choice to attempt any ONE out of the two questions: 5a and 5b.**	
Question Text	Suppose two cards are selected at random without replacement from an ordinary deck of 52 playing cards in the following sequence. A = a Queen is selected, B = a spade is selected Find the probability of $P(B\|A)$
SLO No.	14.4.9
SLO Text	Calculate conditional probability $P(B\|A)$ or $P(A\|B)$, where A and B are dependent events. $P(A \cap B) = P(B) \times P(A \mid B)$
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for finding $P(A)$ 1 mark for finding $P(B)$ 1 mark for finding $P(A \cap B)$ 1 mark for finding $P(B\|A)$
Overall Performance	Lesser number of candidates attempted this question. Candidates who performed well had a strong grasp on the concept of conditional probability and correctly applied the principles to solve the problem. Weaker responses highlighted a misunderstanding of the concept, particularly in handling dependent events and applying the correct probability formulas.
Description of Better Responses	Better responses demonstrated a clear understanding of conditional probability. They correctly applied the formula $P(B\|A) = P(A \cap B)/P(A)$ where A is the event of drawing a Queen and B is the event of drawing a spade. They calculated both the cases where the first card was queen and another one was not the queen, and also the first one if spade's queen so

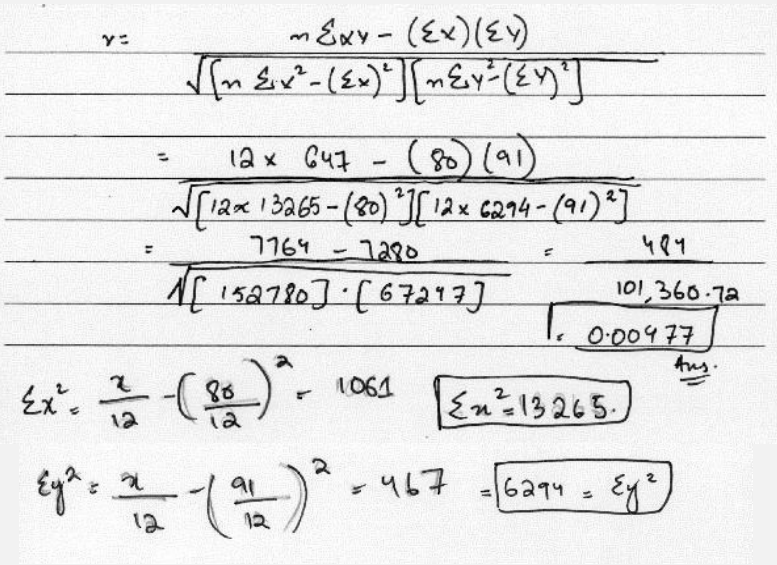
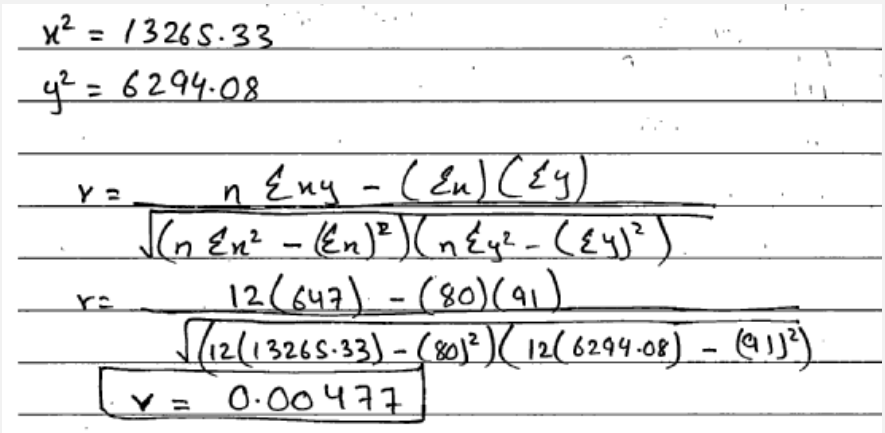
	the next card was from 12 left spades. Such candidates added both the cases and reached at the correct answer.
Image of Better Response	
Description of Weaker Responses	In weaker responses, candidates struggled with understanding the concept of conditional probability. Common issues included: **Errors in Formula Application:** Some candidates applied incorrect formulas or misunderstood the concept of conditional probability. **Incorrect Calculation:** Many candidates failed to correctly find the conditional probability. Some computed the probability of drawing a Queen or a spade independently without considering the dependence of events, leading to incorrect answers. **Incorrect Intersection Calculation:** Many candidates found the probability of intersections incorrectly or applied the formula for independent events, which does not apply to this dependent situation.
Images of Weaker Responses	**Image (i)**  **Image (ii)**  **Image (iii)** 

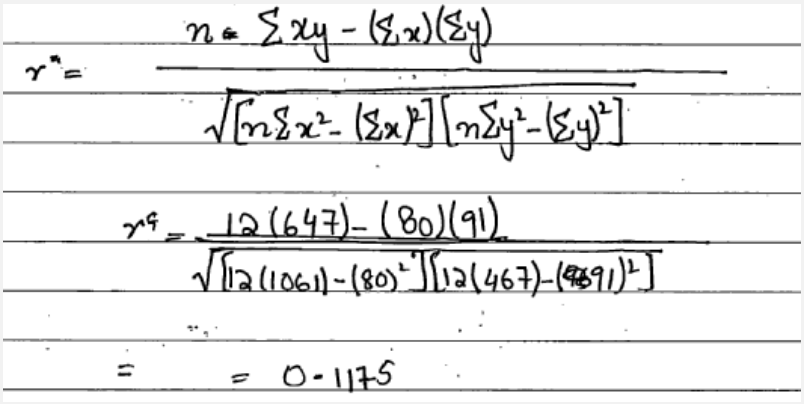
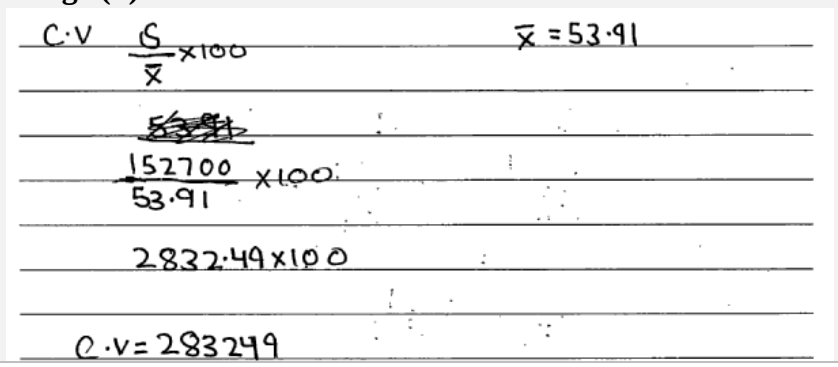
Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:
To enhance the performance in this concept teachers may consider the following strategies;
I**nteractive Probability Simulations:** Use online simulators to demonstrate the effect of drawing cards on probabilities in real-time, helping candidates to visualise the impact of drawing cards without replacement.
Conditional Probability Calculators: Provide tools for candidates to practice and check their calculations, reinforcing their understanding of conditional probability.
Card Games: Use examples from card games to explain conditional probability. For instance, discuss how the probability of drawing certain types of cards changes with each draw.
Practical Scenarios: Incorporate other scenarios involving conditional probabilities, such as drawing marbles from a bag, to make the concept more relatable.

Question No. 6	
Question Text	The relationship between two data sets X and Y is being investigated. Both data sets contain 12 observations each. Their results are as followed: $\sum XY = 647,\ \sum X = 80,\ \sum Y = 91,\ \sigma_x^{\ 2} = 1061,\ \sigma_y^{\ 2} = 467$ Compute the coefficient of correlation between the two variables.
SLO No.	15.2.3
SLO Text	Calculate the coefficient of correlation by using the formula. $r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{\left[n\sum X^2 - (\sum X)^2\right]\left[n\sum Y^2 - (\sum Y)^2\right]}}$
Max Marks	3
Cognitive Level	A
Checking Hints	1 mark for substituting all values correctly 1 mark for finding σ_X & σ_Y 1 mark for finding r
Overall Performance	Candidates overall performed well in this question. Candidates who performed well were able to identify and calculate the necessary components for the coefficient of correlation

	and apply the formula correctly. Weaker responses often showed confusion about the components required and the formula to use, leading to incorrect results.
Description of Better Responses	Candidates who performed well demonstrated a clear understanding of the formula for the coefficient of correlation and the components required for its calculation. They correctly identified that the sums $\sum x^2$ and $\sum y^2$ were necessary for the calculation. Most of the candidates: **Identified Missing Components:** Recognised the need to calculate $\sum x^2$ and $\sum y^2$ from the given data or variances. **Applied the Formula Correctly:** Transformed the correlation formula appropriately and substituted the calculated values accurately. **Correct Calculation:** Followed the correct steps to compute the coefficient of correlation, ensuring accurate results by using the right formula and values.
Images of Better Responses	**Image (i)**  $r = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}}$ $= \frac{12 \times 647 - (80)(91)}{\sqrt{[12 \times 13265 - (80)^2][12 \times 6294 - (91)^2]}}$ $= \frac{7764 - 7280}{\sqrt{[152780] \cdot [67297]}} = \frac{484}{101,360.72}$ $= 0.00477$ Ans. $\Sigma x^2 = \frac{x}{12} - \left(\frac{80}{12}\right)^2 = 1061$ $\Sigma x^2 = 13265.$ $\Sigma y^2 = \frac{x}{12} - \left(\frac{91}{12}\right)^2 = 467 = 6294 = \Sigma y^2$ **Image (ii)**  $x^2 = 13265.33$ $y^2 = 6294.08$ $r = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{(n\Sigma x^2 - (\Sigma x)^2)(n\Sigma y^2 - (\Sigma y)^2)}}$ $r = \frac{12(647) - (80)(91)}{\sqrt{(12(13265.33) - (80)^2)(12(6294.08) - (91)^2)}}$ $r = 0.00477$
Description of Weaker Responses	In weaker responses, candidates faced several challenges: **Misidentification of Components:** Many candidates did not identify that $\sum x^2$ and $\sum y^2$ were missing from their calculations. Instead, they substituted only the variances into the formula for the coefficient of correlation, leading to incorrect results. **Incorrect Formula Application:** Some of the candidates used the incorrect formula for the coefficient of variation or other unrelated formulas, reflecting a misunderstanding of the required calculation. **Misunderstanding of the Requirement:** Incorrect substitution and calculation errors were prevalent, with some candidates calculating the coefficient of variation instead of correlation, indicating a fundamental misunderstanding of the problem.

Images of Weaker Responses	**Image (i)**  $r = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}}$ $r = \frac{12(647) - (80)(91)}{\sqrt{[12(1061) - (80)^2][12(467) - (91)^2]}}$ $= = 0.1175$ **Image (ii)**  C.V $\frac{S}{\bar{x}} \times 100$ $\bar{x} = 53.91$ $\frac{152700}{53.91} \times 100$ 2832.49×100 C.V = 283249

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:
Teachers may apply the following strategies to improve scores in this concept;
Interactive Correlation Calculators: Use online tools that allow students to input data and compute the coefficient of correlation, helping them to apply the formula and understand the process.
Formula Practice Apps: Encourage the use of apps that provide practice problems and feedback on using correlation formulas and calculations.
Practical Data Analysis: Use real-life data sets for correlation analysis, such as student performance and study hours, to make the concept more tangible.
Case Studies: Incorporate case studies that involve correlation calculations to demonstrate practical applications and reinforce understanding.

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.
Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.
Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

Acknowledgements

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Additionally, we express our gratitude to the esteemed team of reviewers for their constructive feedback on overall performance, better and weaker responses, and validating teaching pedagogies along with suggestions for improvement.
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